Summary

Secondary growth deals with the formation of additional vascular tissue by the activities of vascular and cork cambia and secondary thickening meristem (STM). It increases the girth of stem and roots of gymnosperms, most angiosperms, and some monocot plants. Vascular cambium possesses two kinds of initials they are, fusiform and ray initials. Fusiform initials give rise to the axial tissue system whereas ray initials give rise to radial tissue system of stems and roots.

Wood is a very important product of secondary growth. It is classified into various types. Based on respectively on the presence or absence of vessels, - porous and non-porous wood. Based on the wood formed during seasons, - spring wood and autumn wood. The spring and autumn wood, together is called **annual ring**. The lumen of the xylem vessels of heart wood are blocked by many balloon like ingrowths from neighbouring parenchymatous cells called **tyloses**.

The periderm, a secondary protective tissue consists of phellem, phellogen and phelloderm. Secondary growth produces a corky bark around the tree trunk that protects the interior parts from heat, cold, infection etc. Secondary growth of root is different from stem in the method of formation of vascular cambium.

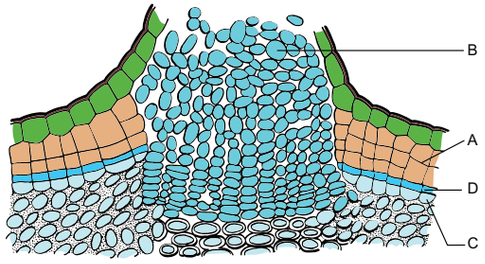
Anomalous secondary growth is now referred as cambial variants. They are abnormal types of secondary growth that occur in some dicots and monocots. Timber is derived from wood logs. In order to enrich the quality of timber, seasoning of wood is done through air and kiln drying. Wood is characterized by colour, grain, texture and figure.

Evaluation

1. Consider the following statements
   In spring season vascular cambium
   i. is less active
   ii. produces a large number of xylary elements
   iii. forms vessels with wide cavities of these,
   a. (i) is correct but (ii) and (iii) are not correct
   b. (i) is not correct but (ii) and (iii) are correct
   c. (i) and (ii) are correct but (iii) is not correct
   d. (i) and (ii) are not correct but (iii) is correct.

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2. Usually, the monocotyledons do not increase their girth, because
   a. They possess actively dividing cambium
   b. They do not possess actively dividing cambium
   c. Ceases activity of cambium
   d. All are correct
3. In the diagram of lenticel identify the parts marked as A,B,C,D



   a. A. phellem, B. Complementary tissue, C. Phelloderm, D. Phellogen.
   b. A. Complementary tissue, B. Phellem, C. Phellogen, D. Phelloderm.
   c. A. Phellogen, B. Phellem, C. Phelloderm, D. complementary tissue
   d. A. Phelloderm, B. Phellem, C. Complementary tissue, D. Phellogen
4. Inner, darker & harder portion of secondary xylem that cannot conduct water in an older dicot stem is called

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